PART 1

2

EXPERIMENT 2

Identification of the Type of Reaction

Aim

To identify the type of reaction.

1. (A) Combustion of magnesium strip in air.
2. (B) Action of dilute sulphuric acid on zinc.
3. (C) Effect of heat on lead nitrate.

Requirements

A burner or spirit lamp, matchbox, candle, splinter, a hard glass test tube, ordinary test tubes, a test tube holder, a pair of tongs, a beaker, sandpaper, a watchglass or china dish, and a glass rod.

Chemicals Required

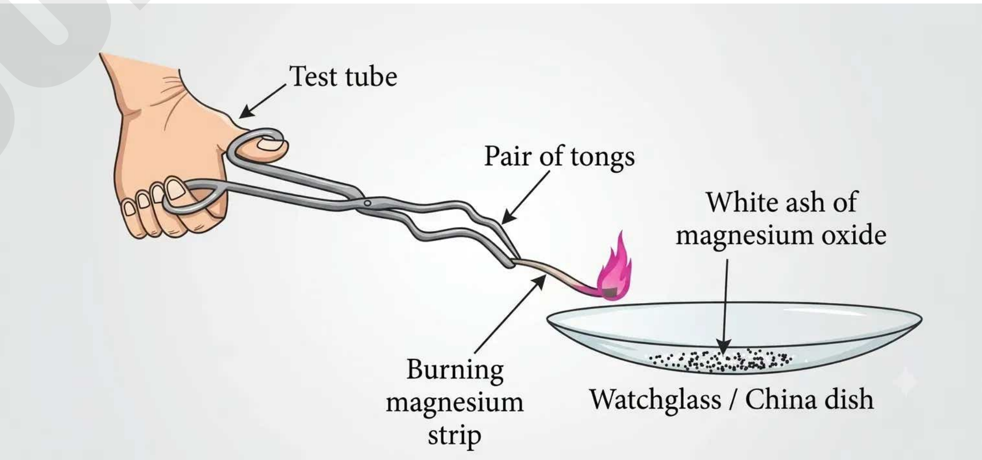
Magnesium ribbon, zinc dust, dilute sulphuric acid, lead nitrate powder, and blue litmus paper.

(A) Combustion of Magnesium Strip in Air

Procedure

1. Hold a strip of magnesium with a pair of tongs and bring it close to the flame of a burner.
2. Collect the ash that remains after burning in a watchglass or china dish.
3. Note the type of reaction taking place, record what you observe, and write down the chemical equation.

Diagram



Observation Table

Reaction / Test	Magnesium strip is burnt. $2Mg + O_2 \rightarrow 2MgO$
Number of reactants	2
Number of products	1
Features of the reaction	There are two reactants and a single product. Magnesium burns with a bright flame. This is a chemical change.
Type of reaction	Combination reaction.

Chemical Reaction

$2Mg\ (s) + O_2\ (g) \rightarrow 2MgO\ (s)$

(Magnesium strip) + (Oxygen) → (Magnesium oxide)

Observations

1. The magnesium strip burns with a luminous flame.
2. A white coloured powder is left behind.

Inference

Since two reactants combine to form a single product, this reaction is a Combination reaction. As a new substance, MgO, is formed, this change is a chemical change.

Precautions

1. Hold the magnesium strip only with a pair of tongs.
2. Wear sunglasses to protect your eyes from the dazzling light of the burning magnesium strip.

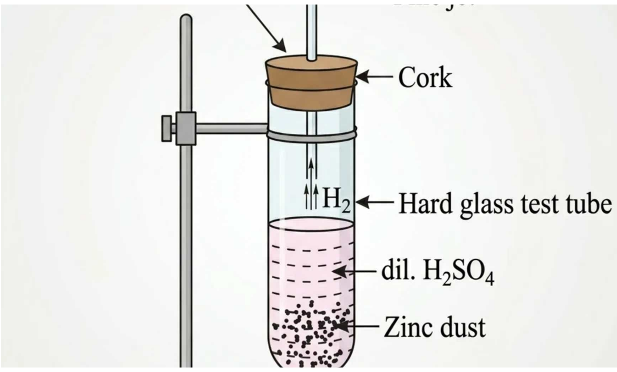
(B) Action of Dilute Sulphuric Acid on Zinc

Procedure

1. Fix a large hard glass test tube onto a clamp stand.
2. Place some zinc dust into the test tube.
3. Pour 3 to 4 ml of dilute sulphuric acid into the test tube so that the zinc dust is completely immersed.

4. Immediately close the mouth of the test tube with a cork fitted with a delivery tube that ends in a fine jet.
5. Bring a lighted candle close to the mouth of the fine jet.
6. Note down your observations, identify the type of reaction, and write the chemical equation.

Diagram



Observation Table

Reaction / Test	Dilute sulphuric acid is added to zinc. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
Number of reactants	2
Number of products	2
Features of the reaction	There are 2 reactants and 2 products. Hydrogen from H_2SO_4 gets displaced by zinc metal.
Type of reaction	Displacement reaction. This is a chemical change.

Chemical Reaction

$Zn\ (s) + H_2SO_4\ (l) \rightarrow ZnSO_4\ (s) + H_2\ (g)$

(Zinc) + (Dil. sulphuric acid) → (Zinc sulphate) + (Hydrogen gas)

Observations

1. The reaction proceeds rapidly, and a colourless gas is produced.
2. The gas produces a pop sound as it burns instantly.
3. The zinc dust disappears completely, leaving behind a colourless solution.

Inferences

1. A reaction in which an atom or group of atoms of one substance is replaced by an atom or group of atoms of another substance is called a displacement reaction.
2. Hydrogen gas is released during this reaction.
3. Since zinc is more reactive than hydrogen, it displaces the less reactive hydrogen from the sulphuric acid, producing a colourless solution.

Precautions

1. Handle the acid with great care.
2. Avoid inhaling the gas that is released.
3. Use a fine jet to observe the burning of hydrogen, as it burns with a mild explosion.

(C) Effect of Heat on Lead Nitrate [$Pb(NO_3)_2$]

Procedure

1. Place a small quantity of white lead nitrate powder into a dry test tube and hold it using a test tube holder.
2. Heat the test tube directly over the burner flame.
3. Hold a moist strip of blue litmus paper at the mouth of the test tube.
4. Bring a glowing splinter near the mouth of the test tube.
5. Record what you observe.
6. Determine the type of reaction and write down the chemical equation.

Observation Table

Reaction	Lead nitrate is heated. $2Pb(NO_3)_2 \xrightarrow{Heat} PbO + 4NO_2\uparrow + O_2\uparrow$
Number of reactants	1
Number of products	3
Features of the reaction	Simple substances are formed. This is a chemical change.
Type of reaction	Decomposition reaction.

Chemical Reaction

$$2Pb(NO_3)_2\ (s) \xrightarrow{Heat} 2PbO\ (s) + 4NO_2\ (g)\uparrow + O_2\ (g)\uparrow$$

(Lead nitrate) → (Lead oxide) + (Nitrogen dioxide) + (Oxygen)

Inferences

1. A reaction in which a single compound breaks down into two or more simpler substances is called a decomposition reaction, for example, the heating of lead nitrate.
2. On heating lead nitrate, nitrogen dioxide (NO_2) and oxygen (O_2) gases are released, leaving behind a yellow solid residue of lead oxide (PbO).
3. Since three simple substances are formed from a single compound, this reaction is classified as a Decomposition reaction.
4. As the nitrogen dioxide gas released is acidic, it turns moist blue litmus red. Since the oxygen gas released supports combustion, the glowing splinter continues to glow.

Precaution

Avoid inhaling the gases released during this reaction.

Multiple Choice Questions

1. Magnesium burns in air to produce ....

(A) $Mg(OH)_2$ (B) MgO (C) MgO_2 (D) $MgCl_2$

Answer: (B) MgO

2. The reaction $CuSO_4$ (aq) + Zn (s) → $ZnSO_4$ (aq) + Cu (s) is a ...... reaction.

(A) displacement (B) double displacement (C) decomposition (D) combination

Answer: (A) displacement

3. When lead nitrate is heated a yellow solid residue of ...... is obtained.

(A) Lead dioxide (B) Lead oxide (C) Lead trioxide (D) Lead sulphate

Answer: (B) Lead oxide

Remember

Chemical reactions involve the breaking and making of bonds between atoms to produce a new substance.

Date: ______________ Teacher's Signature: ______________